

Introduction to Coding theory

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We cover the following topics in this note.

- TBA.

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1 Linear Codes

The fundamental problem of coding theory is to transmit information reliably across a noisy channel. A sender begins with a message, usually represented as a vector

$$\mathbf{x} = (x_1, \dots, x_k) \in \mathbb{F}_2^k,$$

and applies an *encoder*

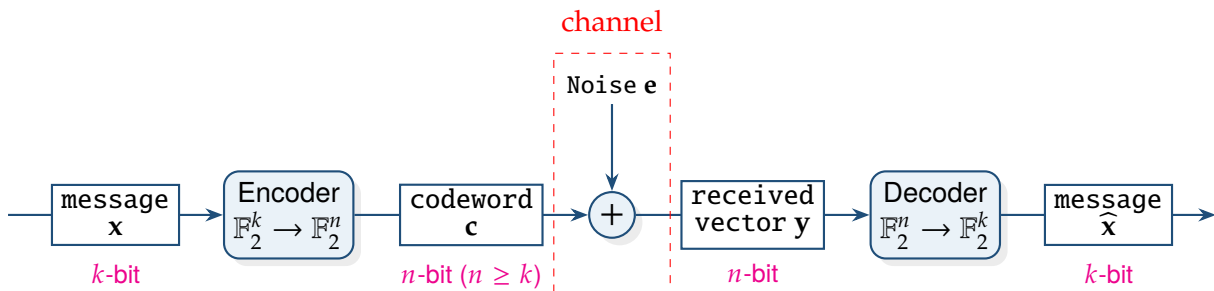
$$\text{Encode} : \mathbb{F}_2^k \longrightarrow \mathbb{F}_2^n, \quad \mathbf{x} \longmapsto \mathbf{c},$$

where $n \geq k$. The vector $\mathbf{c}$ is called the *codeword*.

During transmission, the codeword passes through a noisy channel. The effect of the channel is modeled by adding an error vector $\mathbf{e} \in \mathbb{F}_2^n$ to the transmitted codeword, so that the receiver obtains

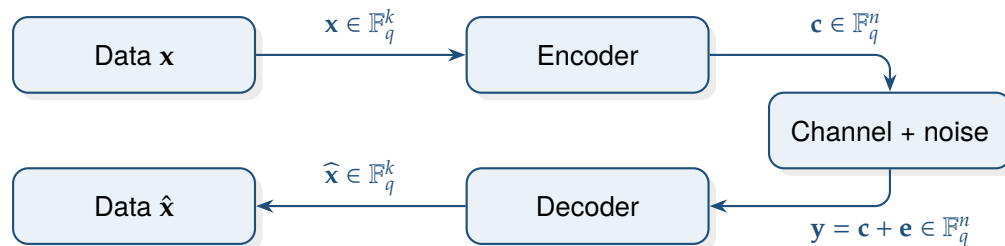
$$\mathbf{y} = \mathbf{c} \oplus \mathbf{e} \in \mathbb{F}_2^n.$$

The vector $\mathbf{y}$ is called the *received vector*. The purpose of the decoder is to recover the original message, or equivalently the transmitted codeword, from the corrupted vector $\mathbf{y}$.



A generic communication model has the form

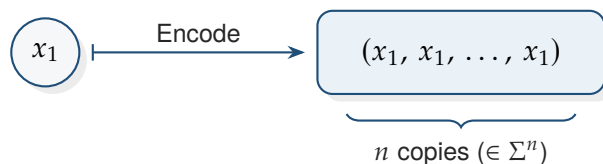
$$\mathbf{x} = (x_1, \dots, x_k) \xrightarrow{\text{encoder}} \underbrace{\mathbf{c} = (c_1, \dots, c_n)}_{\text{codeword } (n \geq k)} \xrightarrow{\text{noise } \mathbf{e}} \mathbf{y} = \mathbf{c} + \mathbf{e} \xrightarrow{\text{decoder}} \hat{\mathbf{x}} = (\hat{x}_1, \dots, \hat{x}_k).$$



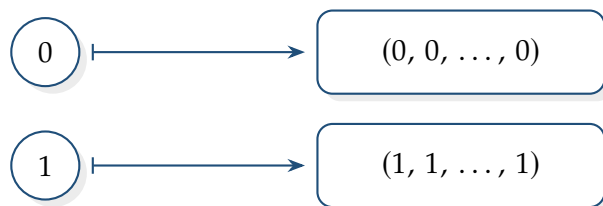
Example 1 (Repetition code). Let Σ be an alphabet. An *encoding* is a map

$$\begin{aligned} \text{Encode} : \quad \Sigma^k &\longrightarrow \Sigma^n \\ (x_1, \dots, x_k) &\longmapsto (c_1, \dots, c_n) \end{aligned}$$

with $n \geq k$.



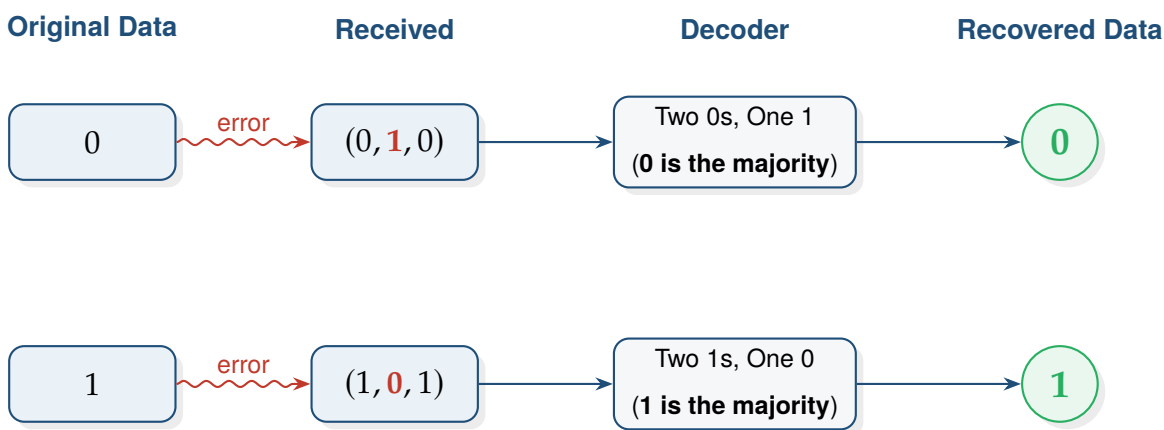
When $k = 1$, a very simple encoding repeats the same symbol



This is called the *repetition code*. If one symbol is changed during transmission, we can still recover the original message by *majority vote*. For instance,

$$(0, 0, 0) \rightsquigarrow (0, 1, 0), \quad (1, 1, 1) \rightsquigarrow (1, 0, 1).$$

Since 0 appears most often in $(0, 1, 0)$, we decode it as 0, and since 1 appears most often in $(1, 0, 1)$, we decode it as 1.



1.1 Definition of Linear Codes

Linear Encoding

Definition 1. A **linear encoding** over $\mathbb{F}_q$ is a linear map

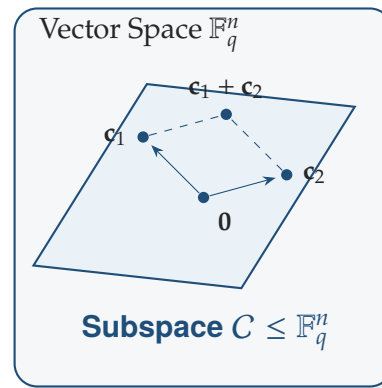
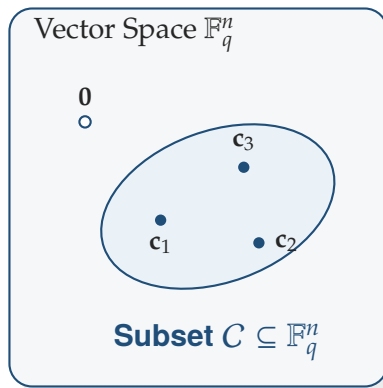
$$\mathbb{F}_q^k \longrightarrow \mathbb{F}_q^n, \quad \mathbf{x} \longmapsto \mathbf{c}.$$

Linear Code

Definition 2. A **code of length** $n \in \mathbb{Z}_{>0}$ **over** $\mathbb{F}_q$ is a subset

$$C \subseteq \mathbb{F}_q^n.$$

The elements of C are called **codewords**. If $C \leq \mathbb{F}_q^n$, then C is called a **linear code**.



Remark 1. If $\dim_{\mathbb{F}_q}(C) = k$, then C is called a $[n, k]_q$ -**code over** $\mathbb{F}_q$.

Proposition 1. If C is an $[n, k]_q$ -code, then

$$|C| = q^k.$$

Proof. Choose a basis $\mathbf{b}_1, \dots, \mathbf{b}_k$ of C . Every codeword can be written uniquely as

$$x_1 \mathbf{b}_1 + \dots + x_k \mathbf{b}_k, \quad x_i \in \mathbb{F}_q.$$

Each x_i has q choices, and there are k independent coefficients. Therefore the number of codewords is q^k . □

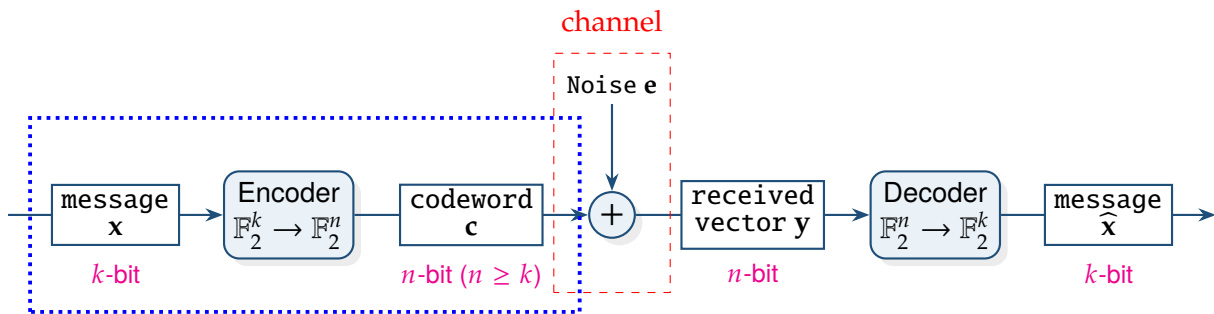
1.2 Generator Matrices

Note. In linear algebra, a choice of bases for finite-dimensional vector spaces V and W identifies each linear transformation $T : V \rightarrow W$ with a matrix.

Accordingly, in coding theory, after choosing bases for $\mathbb{F}_q^k$ and $\mathbb{F}_q^n$, a linear encoding

$$\text{Encode} : \mathbb{F}_q^k \rightarrow \mathbb{F}_q^n$$

is represented by a matrix $\mathbf{G} \in \mathbb{F}_q^{n \times k}$.

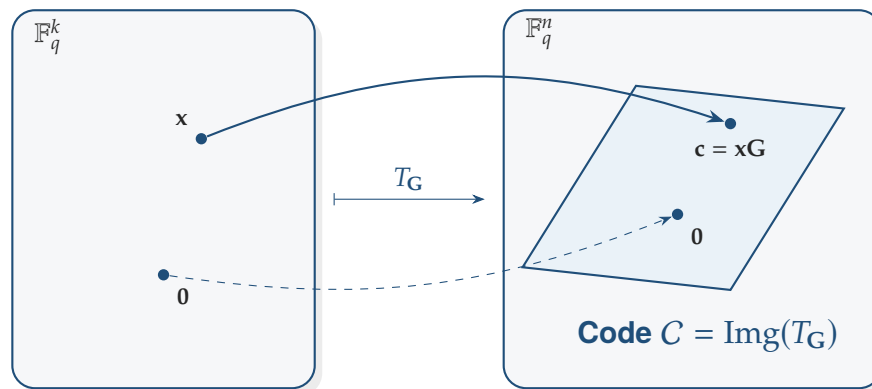


Generator Matrix

Definition 3. Let C be an $[n, k]_q$ -code over $\mathbb{F}_q$. A **generator matrix for C** is a $k \times n$ matrix $\mathbf{G}$ whose rows form a basis of C .

Remark 2. If $\mathbf{x} = (x_1, \dots, x_k) \in \mathbb{F}_q^k$, then encoding is given by

$$\mathbf{c} = \mathbf{x}\mathbf{G} \in \mathbb{F}_q^n \quad \text{with } \mathbf{G} \in \mathbb{F}_q^{k \times n}.$$

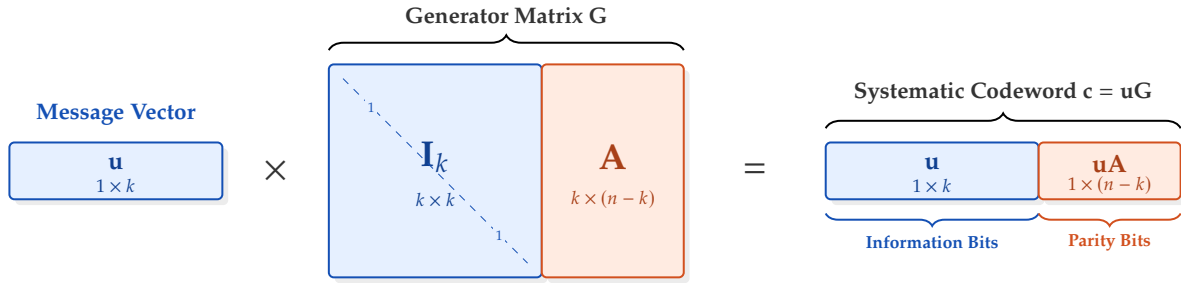


1.2.1 Systematic form

A particularly convenient generator matrix has the form

$$\mathbf{G} = [\mathbf{I}_k \mid \mathbf{A}],$$

where $\mathbf{I}_k$ is the $k \times k$ identity matrix and $\mathbf{A}$ is a $k \times (n - k)$ matrix.



Systematic Form

Definition 4. A code is said to be in **systematic form** if it has a generator matrix of the form

$$\mathbf{G} = [\mathbf{I}_k \mid \mathbf{A}]$$

Remark 3. In general, a linear code has many generator matrices, since different choices of a basis of the code give different matrices. Thus a generator matrix is not unique.

However, if the code is written in *systematic form* with the first k coordinate positions chosen as the information positions, then there is a unique generator matrix of the form $\mathbf{G} = [\mathbf{I}_k \mid \mathbf{A}]$.

Lemma 2. Let $\mathbf{G}, \mathbf{G}' \in \mathbb{F}_q^{k \times n}$ be generator matrices of the same linear code $C \leq \mathbb{F}_q^n$. Then there exists a matrix

$$\exists \mathbf{U} \in GL_k(\mathbb{F}_q) \text{ such that } \mathbf{G}' = \mathbf{UG}.$$

Proof. Let

$$\mathbf{G} = \begin{pmatrix} - & \mathbf{g}_1 & - \\ - & \mathbf{g}_2 & - \\ & \vdots & \\ - & \mathbf{g}_k & - \end{pmatrix}, \quad \mathbf{G}' = \begin{pmatrix} - & \mathbf{g}'_1 & - \\ - & \mathbf{g}'_2 & - \\ & \vdots & \\ - & \mathbf{g}'_k & - \end{pmatrix} \in \mathbb{F}_q^{k \times n}$$

be generator matrices of the same linear code $C \leq \mathbb{F}_q^n$. Then their rows are two bases of the same k -dimensional subspace $C \leq \mathbb{F}_q^n$, so each row of $\mathbf{G}'$ is a linear combination of the rows of $\mathbf{G}$. Thus,

there exist scalars $u_{ij} \in \mathbb{F}_q$ such that

$$\begin{aligned} \mathbf{g}'_1 &= u_{11}\mathbf{g}_1 + u_{12}\mathbf{g}_2 + \cdots + u_{1k}\mathbf{g}_k, \\ \mathbf{g}'_2 &= u_{21}\mathbf{g}_1 + u_{22}\mathbf{g}_2 + \cdots + u_{2k}\mathbf{g}_k, \\ &\vdots \\ \mathbf{g}'_k &= u_{k1}\mathbf{g}_1 + u_{k2}\mathbf{g}_2 + \cdots + u_{kk}\mathbf{g}_k. \end{aligned}$$

Let $\mathbf{U} = (u_{ij}) \in \mathbb{F}_q^{k \times k}$. Then

$$\mathbf{UG} = \begin{pmatrix} u_{11} & u_{12} & \cdots & u_{1k} \\ u_{21} & u_{22} & \cdots & u_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ u_{k1} & u_{k2} & \cdots & u_{kk} \end{pmatrix} \begin{pmatrix} - & \mathbf{g}_1 & - \\ - & \mathbf{g}_2 & - \\ & \vdots & \\ - & \mathbf{g}_k & - \end{pmatrix} = \begin{pmatrix} - & (u_{11}\mathbf{g}_1 + \cdots + u_{1k}\mathbf{g}_k) & - \\ - & (u_{21}\mathbf{g}_1 + \cdots + u_{2k}\mathbf{g}_k) & - \\ & \vdots & \\ - & (u_{k1}\mathbf{g}_1 + \cdots + u_{kk}\mathbf{g}_k) & - \end{pmatrix} = \begin{pmatrix} - & \mathbf{g}'_1 & - \\ - & \mathbf{g}'_2 & - \\ & \vdots & \\ - & \mathbf{g}'_k & - \end{pmatrix} = \mathbf{G}'.$$

So $\mathbf{G}' = \mathbf{UG}$. Similarly, since the rows of $\mathbf{G}$ are also linear combinations of the rows of $\mathbf{G}'$,

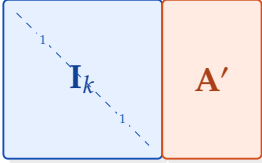
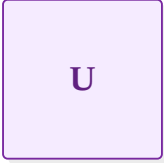
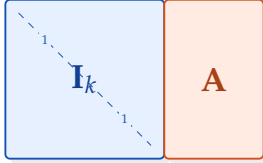
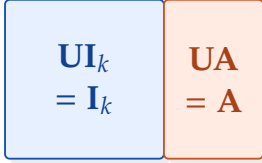
$$\exists \mathbf{V} \in \mathbb{F}_q^{k \times k} \text{ such that } \mathbf{G} = \mathbf{VG}'.$$

Since $\mathbf{G}' = \mathbf{UG}$, we have $\mathbf{G} = \mathbf{VUG}$. Therefore $\mathbf{VU} = \mathbf{I}_k$. Similarly, the equation $\mathbf{G}' = \mathbf{UG} = \mathbf{UVG}$ gives $\mathbf{UV} = \mathbf{I}_k$. Therefore $\mathbf{U}$ is invertible, with inverse $\mathbf{V}$. Hence $\mathbf{U}, \mathbf{V} \in GL_k(\mathbb{F}_q)$. $\square$

Note (Uniqueness of Generator Matrices in Systematic Form). If $\mathbf{G} = [\mathbf{I}_k \mid \mathbf{A}]$ and $\mathbf{G}' = [\mathbf{I}_k \mid \mathbf{A}']$ are two generator matrices for the same code, then $\mathbf{G}' = \mathbf{UG}$ for some $\mathbf{U} \in GL_k(\mathbb{F}_q)$. Then

$$\mathbf{G}' = \mathbf{UG} \implies [\mathbf{I}_k \mid \mathbf{A}'] = \mathbf{U}[\mathbf{I}_k \mid \mathbf{A}] \implies [\mathbf{I}_k \mid \mathbf{A}'] = [\mathbf{UI}_k \mid \mathbf{UA}]$$

so $\mathbf{U} = \mathbf{I}_k$. Hence $\mathbf{G}' = \mathbf{G}$, and therefore $\mathbf{A}' = \mathbf{A}$.

Matrix $\mathbf{G}' = [\mathbf{I}_k \mid \mathbf{A}']$	Matrix $\mathbf{U} \in GL_k(\mathbb{F}_q)$	Matrix $\mathbf{G} = [\mathbf{I}_k \mid \mathbf{A}]$	Product $[\mathbf{UI}_k \mid \mathbf{UA}]$
	$=$ 	$\times$ 	$=$ 

Remark 4. Let the first k coordinates of the codeword are exactly the information symbols. If $\mathbf{x} = (x_1, \dots, x_k)$, then

$$\mathbf{xG} = (x_1, \dots, x_k, c_{k+1}, \dots, c_n).$$

$$\begin{array}{c} \text{Message Vector} \\ \mathbf{x} = (x_1, \dots, x_k) \\ 1 \times k \end{array} \times \begin{array}{c} \text{Generator Matrix } \mathbf{G} \\ \left[\begin{array}{c|c} \mathbf{I}_k & \mathbf{A} \\ \hline k \times k & k \times (n-k) \end{array} \right] \end{array} = \begin{array}{c} \text{Systematic Codeword } \mathbf{c} = \mathbf{xG} \\ \left(\begin{array}{c|c} x_1, \dots, x_k & c_{k+1}, \dots, c_n \\ \hline 1 \times k & 1 \times (n-k) \end{array} \right) \\ \text{Information Symbols} \quad \text{Parity Symbols} \end{array}$$

So the information symbols appear in the transmitted codeword, followed by the parity symbols.

Note. Let $\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_k \end{pmatrix} \in \mathbb{F}_q^k$ be the message vector, written as a column vector. With the column-vector

convention, the encoding map is represented by a generator matrix $\mathbf{G}$ such that $\mathbf{Gx} = \begin{pmatrix} x_1 \\ \vdots \\ x_k \\ c_{k+1} \\ \vdots \\ c_n \end{pmatrix}$.

$$\begin{array}{c} \text{Generator Matrix } \mathbf{G} \\ \left[\begin{array}{c} \mathbf{I}_k \\ \hline \mathbf{A}^T \\ k \times k \\ (n-k) \times k \end{array} \right] \end{array} \times \begin{array}{c} \text{Message Vector} \\ \mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_k \end{pmatrix} \\ k \times 1 \end{array} = \begin{array}{c} \begin{pmatrix} x_1 \\ \vdots \\ x_k \\ c_{k+1} \\ \vdots \\ c_n \end{pmatrix} \\ \left[\begin{array}{c} \mathbf{x} \\ \hline \mathbf{A}^T \mathbf{x} \\ k \times 1 \\ (n-k) \times 1 \end{array} \right] \end{array}$$

$\leftarrow \text{Info}$
 $\text{Systematic Codeword } \mathbf{c} = \mathbf{Gx}$
 $\leftarrow \text{Parity}$

1.2.2 Examples: Repetition Code and Single Parity-check Code

Example 2 (Repetition code). The $[n, 1]_q$ repetition code is the linear code defined by

$$\text{Encode} : \mathbb{F}_q \longrightarrow \mathbb{F}_q^n, \quad (x_1) \longmapsto (x_1, x_1, \dots, x_1).$$

Its generator matrix is

$$\mathbf{G} = [1 \ 1 \ \dots \ 1]_{1 \times n}.$$

Indeed,

$$(x_1)\mathbf{G} = (x_1)[1 \ 1 \ \dots \ 1] = (x_1, x_1, \dots, x_1).$$

Hence the code is

$$C = \{(a, a, \dots, a) \in \mathbb{F}_q^n : a \in \mathbb{F}_q\} \leq \mathbb{F}_q^n.$$

Example 3 (Single Parity-check Code). The single parity-check code $[n, n-1]_q$ is the linear code

$$C = \left\{ (x_1, \dots, x_{n-1}, \sum_{i=1}^{n-1} x_i) \in \mathbb{F}_q^n : (x_1, \dots, x_k) \in \mathbb{F}_q^k \right\} \leq \mathbb{F}_q^n.$$

Equivalently, it is defined by the encoding map

$$\text{Encode} : \mathbb{F}_q^{n-1} \longrightarrow \mathbb{F}_q^n, \quad (x_1, \dots, x_{n-1}) \longmapsto (x_1, \dots, x_k, \sum_{i=1}^{n-1} x_i).$$

A generator matrix in systematic form is

$$\mathbf{G} = \begin{bmatrix} 1 & 0 & \dots & 0 & 1 \\ 0 & 1 & \dots & 0 & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & 1 \end{bmatrix}_{k \times n} = \begin{bmatrix} & 1 \\ \mathbf{I}_k & \vdots \\ & 1 \end{bmatrix} = [\mathbf{I}_k \mid \mathbf{1}],$$

where $\mathbf{1} \in \mathbb{F}_q^k$ denotes the column vector all of whose entries are 1. Indeed, for $\mathbf{x} = (x_1, \dots, x_k) \in \mathbb{F}_q^k$,

$$\mathbf{x}\mathbf{G} = (x_1, \dots, x_k) \begin{bmatrix} & 1 \\ \mathbf{I}_k & \vdots \\ & 1 \end{bmatrix} = (x_1, \dots, x_k, x_1 + \dots + x_k).$$

Hence the first $n-1$ coordinates are the information symbols, and the last coordinate is their parity symbol.

1.3 Parity-check Matrices

A linear code can be described either by its generators or by the linear equations that all codewords satisfy.

Parity-check Matrix

Definition 5. Let C be an $[n, k]_q$ -code. A matrix $H \in \mathbb{F}_q^{(n-k) \times n}$ is called a **parity-check matrix** for C if

$$C = \{\mathbf{v} \in \mathbb{F}_q^n : H\mathbf{v}^T = 0\}.$$

Thus codewords are exactly the vectors satisfying a system of $n - k$ homogeneous linear equations.

1.3.1 Parity-check Matrix from a Systematic Generator Matrix

Note. Suppose

$$\mathbf{G} = [\mathbf{I}_k \mid \mathbf{A}],$$

is a generator matrix in systematic form, where $\mathbf{A} \in \mathbb{F}_q^{k \times (n-k)}$. Then define

$$\mathbf{H} = [-\mathbf{A}^T \mid \mathbf{I}_{n-k}].$$

Parity-check Matrix of a Systematic Generator Matrix

Proposition 3. Let C be an $[n, k]_q$ -code, and suppose that C is generated by the systematic generator matrix

$$\mathbf{G} = [\mathbf{I}_k \mid \mathbf{A}] \in \mathbb{F}_q^{k \times n},$$

where $\mathbf{A} \in \mathbb{F}_q^{k \times (n-k)}$. Then

$$\mathbf{H} = [-\mathbf{A}^T \mid \mathbf{I}_{n-k}] \in \mathbb{F}_q^{(n-k) \times n}$$

is a parity-check matrix of C .

Proposition 4 (Parity-check matrix of a systematic generator matrix). Let C be an $[n, k]_q$ -code, and suppose that C is generated by the systematic generator matrix

$$G = [I_k \mid A] \in M_{k \times n}(\mathbb{F}_q),$$

where $A \in M_{k \times (n-k)}(\mathbb{F}_q)$. Then

$$H = [-A^T \mid I_{n-k}] \in M_{(n-k) \times n}(\mathbb{F}_q)$$

is a parity-check matrix of C .

Proof. By definition, it suffices to show that

$$GH^{\top} = 0.$$

Now

$$H^{\top} = \begin{bmatrix} -A \\ I_{n-k} \end{bmatrix},$$

so

$$GH^{\top} = [I_k \mid A] \begin{bmatrix} -A \\ I_{n-k} \end{bmatrix} = I_k(-A) + AI_{n-k} = -A + A = 0.$$

□

Note (Linear Code as Short Exact Sequence). A linear $[n, k]_q$ code is most naturally described by a short exact sequence

$$0 \longrightarrow \mathbb{F}_q^k \xrightarrow{\text{Encode}} \mathbb{F}_q^n \xrightarrow{H} \mathbb{F}_q^{n-k} \longrightarrow 0,$$

A message $\mathbf{x} \in \mathbb{F}_q^k$ is encoded as

$$\mathbf{c} = \text{Enc}(\mathbf{x}) \in C.$$

After transmission through the channel, one receives

$$\mathbf{y} = \mathbf{c} + \mathbf{e} \in \mathbb{F}_q^n,$$

where $\mathbf{e} \in \mathbb{F}_q^n$ is the error vector. Applying the parity-check map yields

$$H(\mathbf{y}) = H(\mathbf{c} + \mathbf{e}) = H(\mathbf{e}),$$

since $H(\mathbf{c}) = 0$ for every $\mathbf{c} \in C$.

Thus decoding may be viewed as follows: from the syndrome $H(\mathbf{y})$, choose an error estimate $\hat{\mathbf{e}}$ with the same syndrome, typically of smallest Hamming weight, and define

$$\hat{\mathbf{c}} = \mathbf{y} - \hat{\mathbf{e}} \in C.$$

Then recover the message by

$$\hat{\mathbf{x}} = \text{Enc}^{-1}(\hat{\mathbf{c}}).$$

Hence error correction is the procedure of moving the received word from its ambient coset in $\mathbb{F}_q^n/C$ back to the code $C = \ker(H)$, and then inverting the encoding map.

Cardinality of the matrix space $\text{Mat}_m(\mathbb{F}_q)$

Proposition 5. Let $m, n \in \mathbb{Z}_{>0}$, and let q be a prime power. The number of $m \times n$ matrices over the finite field $\mathbb{F}_q$ is

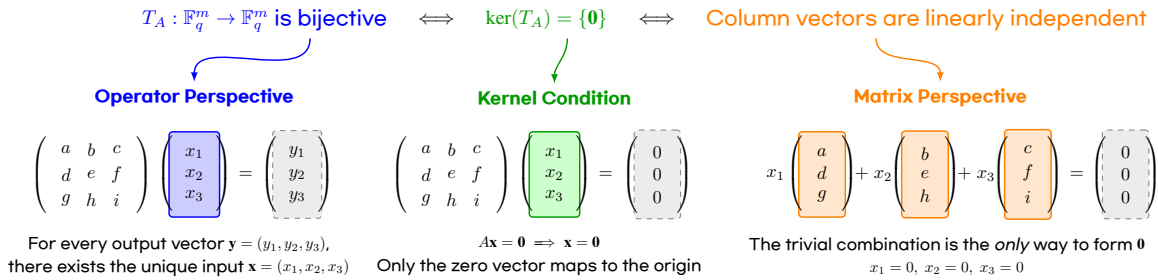
$$|\text{Mat}_{m \times n}(\mathbb{F}_q)| = q^{mn}.$$

In particular, $|\text{Mat}_m(\mathbb{F}_q)| = q^{m^2}$.

Remark 5. Let

$$A = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_m \\ | & | & \cdots & | \end{pmatrix} \in \text{Mat}_{m \times m}(\mathbb{F}_q),$$

where $\mathbf{v}_i \in \mathbb{F}_q^m$ are the columns of A . Then A is invertible if and only if its columns are linearly independent.



Observation (Computation of $|\text{GL}_2(\mathbb{F}_q)|$).

A 2×2 matrix has the form

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad a, b, c, d \in \mathbb{F}_q.$$

Write the columns as

$$\mathbf{v}_1 = \begin{pmatrix} a \\ c \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} b \\ d \end{pmatrix}.$$

$$|\text{GL}_2(\mathbb{F}_q)| = (q^2 - 1) \cdot (q^2 - q)$$

$$\begin{pmatrix} \boxed{a} & b \\ \boxed{c} & d \end{pmatrix} \quad \begin{pmatrix} \boxed{a} & \boxed{b} \\ \boxed{c} & \boxed{d} \end{pmatrix}$$

$$\begin{matrix} \mathbf{v}_1 \neq \mathbf{0} \\ q^2 - 1 \text{ ways} \end{matrix} \quad \times \quad \begin{matrix} \mathbf{v}_2 \notin \text{Span}(\mathbf{v}_1) \\ q^2 - q \text{ ways} \end{matrix}$$

(i) The vector $\mathbf{v}_1$ can be any nonzero vector in $\mathbb{F}_q^2$. Hence there are $q^2 - 1$ choices.

(ii) Since $\text{span}(\mathbf{v}_1) = \{\lambda \mathbf{v}_1 : \lambda \in \mathbb{F}_q\}$ has exactly q elements, there are $q^2 - q$ choices for $\mathbf{v}_2$.

Multiplying these choices, we obtain

$$|\text{GL}_2(\mathbb{F}_q)| = (q^2 - 1)(q^2 - q).$$

Observation (Computation of $|GL_3(\mathbb{F}_q)|$).

$$|GL_3(\mathbb{F}_q)| = (q^3 - 1) \cdot (q^3 - q) \cdot (q^3 - q^2)$$

$$\begin{pmatrix} \boxed{a} & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \quad \begin{pmatrix} \boxed{a} & \boxed{b} & c \\ d & e & f \\ g & h & i \end{pmatrix} \quad \begin{pmatrix} \boxed{a} & b & \boxed{c} \\ d & e & f \\ g & h & i \end{pmatrix}$$

$\mathbf{v}_1 \neq \mathbf{0}$ $\times$ $\mathbf{v}_2 \notin \text{Span}(\mathbf{v}_1)$ $\times$ $\mathbf{v}_3 \notin \text{Span}(\mathbf{v}_1, \mathbf{v}_2)$
 $q^3 - 1$ ways $q^3 - q$ ways $q^3 - q^2$ ways

Observation (Computation of $|GL_4(\mathbb{F}_q)|$).

$$|GL_4(\mathbb{F}_q)| = (q^4 - 1) \cdot (q^4 - q) \cdot (q^4 - q^2) \cdot (q^4 - q^3)$$

$$\begin{pmatrix} \boxed{a} & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{pmatrix} \quad \begin{pmatrix} \boxed{a} & \boxed{b} & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{pmatrix} \quad \begin{pmatrix} \boxed{a} & b & \boxed{c} & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{pmatrix} \quad \begin{pmatrix} \boxed{a} & b & c & \boxed{d} \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{pmatrix}$$

$\mathbf{v}_1 \neq \mathbf{0}$ $\times$ $\mathbf{v}_2 \notin \text{Span}(\mathbf{v}_1)$ $\times$ $\mathbf{v}_3 \notin \text{Span}(\mathbf{v}_1, \mathbf{v}_2)$ $\times$ $\mathbf{v}_4 \notin \text{Span}(\mathbf{v}_i)_{i=1}^3$
 $q^4 - 1$ ways $q^4 - q$ ways $q^4 - q^2$ ways $q^4 - q^3$ ways

Counting invertible matrices over a finite field $\mathbb{F}_q$

Theorem 6. Let $m, n \in \mathbb{Z}_{>0}$, and let q be a prime power. The number of invertible $m \times m$ matrices over $\mathbb{F}_q$ is

$$|GL_m(\mathbb{F}_q)| = (q^m - 1)(q^m - q)(q^m - q^2) \cdots (q^m - q^{m-1}).$$

Equivalently, $|GL_m(\mathbb{F}_q)| = \prod_{i=0}^{m-1} (q^m - q^i).$

Remark 6. Recall that

$$|\text{Mat}_m(\mathbb{F}_q)| = q^{m^2}, \quad \text{and} \quad |GL_m(\mathbb{F}_q)| = \prod_{i=0}^{m-1} (q^m - q^i),$$

and hence

$$\begin{aligned}
 \frac{|GL_m(\mathbb{F}_q)|}{|\text{Mat}_m(\mathbb{F}_q)|} &= \frac{\prod_{i=0}^{m-1} (q^m - q^i)}{q^{m^2}} = \frac{1}{q^{m^2}} \left[(q^m - 1)(q^m - q)(q^m - q^2) \cdots (q^m - q^{m-1}) \right] \\
 &= \frac{1}{q^{m^2}} \left[q^m \left(1 - \frac{1}{q^m}\right) \left(1 - \frac{1}{q^{m-1}}\right) \left(1 - \frac{1}{q^{m-2}}\right) \cdots \left(1 - \frac{1}{q^1}\right) \right] \\
 &= \left(1 - \frac{1}{q^m}\right) \left(1 - \frac{1}{q^{m-1}}\right) \left(1 - \frac{1}{q^{m-2}}\right) \cdots \left(1 - \frac{1}{q^1}\right) \\
 &= \prod_{j=1}^m (1 - q^{-j}).
 \end{aligned}$$

m	$ \text{Mat}_m(\mathbb{F}_q) $	$ GL_m(\mathbb{F}_q) $	$\frac{ GL_m(\mathbb{F}_q) }{ \text{Mat}_m(\mathbb{F}_q) }$
1	q	$q - 1$	$1 - q^{-1}$
2	q^4	$(q^2 - 1)(q^2 - q)$	$(1 - q^{-1})(1 - q^{-2})$
3	q^9	$(q^3 - 1)(q^3 - q)(q^3 - q^2)$	$(1 - q^{-1})(1 - q^{-2})(1 - q^{-3})$
4	q^{16}	$(q^4 - 1)(q^4 - q)(q^4 - q^2)(q^4 - q^3)$	$(1 - q^{-1})(1 - q^{-2})(1 - q^{-3})(1 - q^{-4})$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
m	q^{m^2}	$\prod_{i=0}^{m-1} (q^m - q^i)$	$\prod_{j=1}^m (1 - q^{-j})$

Example 4 (Invertible binary matrices). For $q = 2$, we have

$$|\text{Mat}_m(\mathbb{F}_2)| = 2^{m^2}, \quad \frac{|GL_m(\mathbb{F}_2)|}{|\text{Mat}_m(\mathbb{F}_2)|} = \prod_{j=1}^m (1 - 2^{-j}).$$

m	$ \text{Mat}_m(\mathbb{F}_2) $	$ GL_m(\mathbb{F}_2) $	$\frac{ GL_m(\mathbb{F}_2) }{ \text{Mat}_m(\mathbb{F}_2) }$
1	2	1	0.500000
2	16	6	0.375000
3	512	168	0.328125
4	65,536	20,160	0.307617
$\vdots$	$\vdots$	$\vdots$	$\vdots$
20	—	—	0.288788

```

q = 2

print(f"{'m':>2} {'|Mat_m(F_q)|':>15} {'|GL_m(F_q)|':>15} {'ratio':>12}")
print("-" * 48)

for m in range(1, 7):
    Mat = q^(m^2)
    GL = prod(q^m - q^i for i in range(m))
    ratio = GL / Mat
    print(f"{'m':>2} {'Mat':>15} {'GL':>15} {'ratio.n():>12.6f}")

```

m	Mat_m(F_q)	GL_m(F_q)	ratio
1	2	1	0.500000
2	16	6	0.375000
3	512	168	0.328125
4	65536	20160	0.307617
5	33554432	9999360	0.298004
6	68719476736	20158709760	0.293348

Proposition 7. Let $q \geq 2$ be fixed, and define

$$a_m := \frac{|GL_m(\mathbb{F}_q)|}{|\text{Mat}_m(\mathbb{F}_q)|} = \prod_{j=1}^m (1 - q^{-j}).$$

Then the sequence $\langle a_m \rangle_{m \geq 1}$ converges as $m \rightarrow \infty$. In other words, there exists $\lim_{n \rightarrow \infty} \prod_{j=1}^m (1 - q^{-j}) \in \mathbb{R}$.

Proof. For every $j \geq 1$,

$$q \geq 2 \implies q^j \geq 2^j > 1 \implies 0 < \frac{1}{q^j} < 1.$$

Therefore

$$0 < \frac{1}{q^j} < 1 \implies -1 < -\frac{1}{q^j} < 0 \implies 0 < 1 - \frac{1}{q^j} < 1.$$

It follows that for each $m \geq 1$,

$$a_m = \prod_{j=1}^m (1 - q^{-j}) > 0.$$

So the sequence $\langle a_m \rangle$ is bounded below by 0. Moreover,

$$a_{m+1} = \prod_{j=1}^{m+1} (1 - q^{-j}) = \left[\prod_{j=1}^m (1 - q^{-j}) \right] \cdot (1 - q^{-(m+1)}) = a_m (1 - q^{-(m+1)}).$$

Since $0 < 1 - q^{-(m+1)} < 1$, we obtain $a_{m+1} < a_m$. Thus $\langle a_m \rangle$ is decreasing. Therefore $\langle a_m \rangle$ is a monotone decreasing sequence which is bounded below. Hence it converges. $\square$

Example 5. For small prime q , the limit is approximately

$$\begin{aligned} q = 2 : & \quad \prod_{j=1}^{\infty} (1 - 2^{-j}) \approx 0.288788095 \\ q = 3 : & \quad \prod_{j=1}^{\infty} (1 - 3^{-j}) \approx 0.560126077 \\ q = 5 : & \quad \prod_{j=1}^{\infty} (1 - 5^{-j}) \approx 0.760332795 \end{aligned}$$

```
q = 2
r = 3
s = 5
R = RealField(100)

L = prod(R(1) - R(q)^(-j) for j in range(1, 100))
M = prod(R(1) - R(r)^(-j) for j in range(1, 100))
N = prod(R(1) - R(s)^(-j) for j in range(1, 100))

print(L)
print(M)
print(N)
```

```
0.28878809508660242127889972193
0.56012607792794894496979224331
0.76033279587123242010148829629
```

1.4 Weights and Distances

2 Cyclic Codes

In this section, we have three goals:

- (1) To understand the ideal correspondence between a ring R and its quotient

$$R \longrightarrow R/I.$$

- (2) To interpret this correspondence geometrically through spectra, first in the familiar examples

$$\mathbb{Z} \quad \text{and} \quad \mathbb{C}[x].$$

- (3) To apply the same idea to the quotient ring

$$\mathbb{F}_q[x]/\langle x^n - 1 \rangle$$

and explain why its ideals are exactly the cyclic codes of length n over $\mathbb{F}_q$.

The central philosophy is the following:

Passing from R to R/I means forcing the elements of I to become zero. Therefore only those ideals of R that already contain I survive in the quotient.

In the cyclic-code setting, the additional feature is that the quotient ring

$$\mathbb{F}_q[x]/\langle x^n - 1 \rangle$$

is naturally identified with the vector space $\mathbb{F}_q^n$, and multiplication by x becomes cyclic shift of coordinates. This turns ring-theoretic ideals into coding-theoretic cyclic codes.

2.1 The ideal correspondence for a quotient ring

Ideal Correspondence

Theorem 8. Let R be a ring, let $I \triangleleft R$ be an ideal, and let

$$\pi : R \longrightarrow R/I$$

be the quotient map. Then the assignment

$$J' \longmapsto \pi^{-1}(J')$$

defines an inclusion-preserving bijection

$$\{\text{ideals of } R/I\} \longleftrightarrow \{\text{ideals } J \triangleleft R \text{ such that } I \subseteq J\}.$$

Its inverse is given by

$$J \longmapsto J/I.$$

Proof. Let $J' \triangleleft R/I$. Then $\pi^{-1}(J')$ is an ideal of R . Since $\pi(I) = 0$, we have

$$I \subseteq \pi^{-1}(J').$$

Conversely, let $J \triangleleft R$ with $I \subseteq J$. Then

$$J/I = \{\bar{x} \in R/I : x \in J\}$$

is an ideal of R/I .

Now we check that these constructions are inverse to one another.

If $J \triangleleft R$ with $I \subseteq J$, then

$$\pi^{-1}(J/I) = J.$$

Indeed, $\pi(x) \in J/I$ if and only if $x \in J$.

If $J' \triangleleft R/I$, then

$$\pi^{-1}(J')/I = J'.$$

Indeed, every element of J' has the form $\bar{x}$ with $x \in \pi^{-1}(J')$.

Thus the correspondence is bijective, and it is plainly inclusion-preserving. $\square$

Remark 7. This theorem says

$$\text{ideals of } R/I \longleftrightarrow \text{ideals of } R \text{ containing } I.$$

2.2 Cyclic codes as ideals

Observation. Consider the code

$$C = \{000, 011, 101, 110\} \leq \mathbb{F}_2^3.$$

Under the isomorphism

$$\begin{aligned} \Phi : \quad \mathbb{F}_2^3 &\longrightarrow \mathbb{F}_2[x]/\langle x^3 - 1 \rangle \\ (c_0, c_1, c_2) &\longmapsto c_0 + c_1x + c_2x^2 \end{aligned}$$

the cyclic shift of coordinates corresponds to multiplication by x modulo $x^3 - 1$.

Codeword	Vector	Polynomial
000	(0, 0, 0)	0
011	(0, 1, 1)	$x + x^2$
101	(1, 0, 1)	$1 + x^2$
110	(1, 1, 0)	$1 + x$

So in polynomial form,

$$C = \{0, x + x^2, 1 + x^2, 1 + x\} \subseteq \mathbb{F}_2[x]/\langle x^3 - 1 \rangle.$$

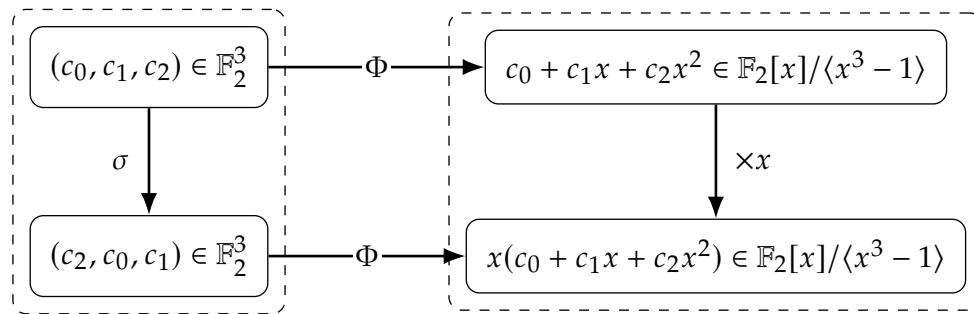
Now observe that

$$x^3 - 1 = x^3 + 1 = (x + 1)(x^2 - x + 1) = (x + 1)(x^2 + x + 1)$$

in $\mathbb{F}_2[x]$.

$C \leq \mathbb{F}_2^3$ closed under cyclic shift

$\Phi(C) \subseteq \mathbb{F}_2[x]/\langle x^3 - 1 \rangle$ closed under $\times x$



$$\sigma(c_0, c_1, c_2) = (c_2, c_0, c_1)$$

$$x(c_0 + c_1x + c_2x^2) \equiv c_2 + c_0x + c_1x^2 \pmod{x^3 - 1}$$

Cyclic code

Definition 6. A linear code $C \leq \mathbb{F}_q^n$ is called a **cyclic code** if

$$c = (c_0, c_1, \dots, c_{n-1}) \in C \implies c^{\ggg 1} = (c_{n-1}, c_0, c_1, \dots, c_{n-2}) \in C.$$

In other words, the cyclic shift of every codeword is also a codeword.

Remark 8. Consider the cyclic shift operator

$$T : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^n, \quad T(c_0, c_1, \dots, c_{n-1}) = (c_{n-1}, c_0, c_1, \dots, c_{n-2}),$$

Then C is cyclic if and only if $T(C) = C$. Thus a cyclic code is a linear subspace of $\mathbb{F}_q^n$ invariant under cyclic shift.

Cyclic codes are Ideals

Theorem 9. Identify $\mathbb{F}_q^n$ with the quotient ring $\mathbb{F}_q[x]/\langle x^n - 1 \rangle$ by

$$(c_0, c_1, \dots, c_{n-1}) \longleftrightarrow c_0 + c_1x + \dots + c_{n-1}x^{n-1} \pmod{x^n - 1}.$$

Under this identification, a linear code $C \leq \mathbb{F}_q^n$ is cyclic if and only if the corresponding subset is an ideal of

$$\mathbb{F}_q[x]/\langle x^n - 1 \rangle.$$

Proof. Under the above identification, multiplication by x in $\mathbb{F}_q[x]/\langle x^n - 1 \rangle$ sends

$$c_0 + c_1x + \dots + c_{n-1}x^{n-1}$$

to

$$c_{n-1} + c_0x + c_1x^2 + \dots + c_{n-2}x^{n-1},$$

because $x^n \equiv 1 \pmod{x^n - 1}$. This is exactly the cyclic shift

$$(c_0, c_1, \dots, c_{n-1}) \mapsto (c_{n-1}, c_0, c_1, \dots, c_{n-2}).$$

Hence a linear subspace $C \leq \mathbb{F}_q^n$ is closed under cyclic shift if and only if the corresponding subset of $\mathbb{F}_q[x]/\langle x^n - 1 \rangle$ is closed under multiplication by x . Since it is already closed under addition and scalar multiplication, this is equivalent to being an ideal. $\square$

Example 6 (The ideal lattice of $\mathbb{F}_2[x]/\langle x^3 - 1 \rangle$ and the corresponding cyclic codes). Let

$$A = \mathbb{F}_2[x]/\langle x^3 - 1 \rangle.$$

Since in $\mathbb{F}_2[x]$ we have

$$x^3 - 1 = x^3 + 1 = (x + 1)(x^2 + x + 1),$$

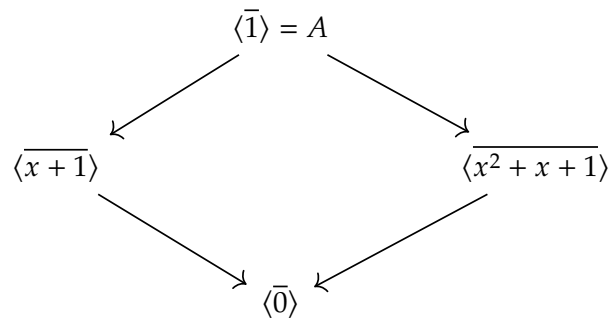
the monic divisors of $x^3 - 1$ are

$$1, \quad x + 1, \quad x^2 + x + 1, \quad x^3 + 1.$$

Therefore the ideals of A are exactly

$$\langle \bar{1} \rangle, \quad \langle \overline{x + 1} \rangle, \quad \langle \overline{x^2 + x + 1} \rangle, \quad \langle \overline{x^3 + 1} \rangle = \langle \bar{0} \rangle.$$

The ideal lattice is therefore



We now describe the corresponding cyclic codes under the identification

$$(c_0, c_1, c_2) \longleftrightarrow c_0 + c_1x + c_2x^2$$

between $\mathbb{F}_2^3$ and A .

1. The ideal

$$\langle \bar{1} \rangle = A$$

corresponds to the whole space

$$\mathbb{F}_2^3 = \{000, 001, 010, 011, 100, 101, 110, 111\}.$$

2. The ideal

$$\langle \overline{x + 1} \rangle$$

is generated by $x + 1$. Its elements are

$$0, \quad x + 1, \quad x(x + 1) = x^2 + x, \quad x^2(x + 1) = x^3 + x^2 \equiv 1 + x^2 \pmod{x^3 - 1}.$$

Thus

$$\langle \overline{x + 1} \rangle = \{0, x + 1, x^2 + x, 1 + x^2\}.$$

In vector form, this is

$$\{000, 110, 011, 101\} = \{000, 011, 101, 110\}.$$

This is the even-weight code of length 3.

3. The ideal

$$\langle \overline{x^2 + x + 1} \rangle$$

is generated by $x^2 + x + 1$. Since

$$x(x^2 + x + 1) = x^3 + x^2 + x \equiv 1 + x^2 + x = x^2 + x + 1 \pmod{x^3 - 1},$$

multiplication by x does not produce a new element. Hence

$$\langle \overline{x^2 + x + 1} \rangle = \{0, x^2 + x + 1\}.$$

In vector form, this is

$$\{000, 111\},$$

the repetition code of length 3.

4. The ideal

$$\langle \overline{0} \rangle$$

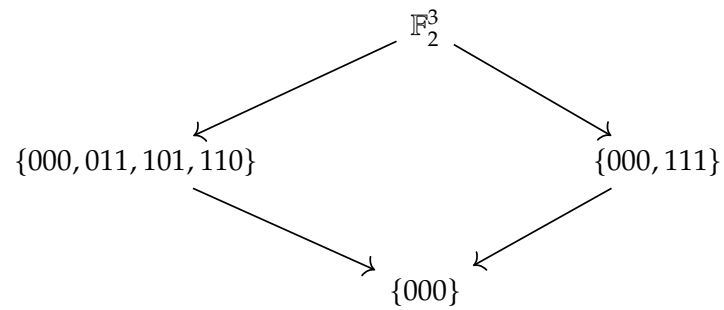
corresponds to the zero code

$$\{000\}.$$

Therefore the four cyclic codes of length 3 over $\mathbb{F}_2$ are exactly

$$\mathbb{F}_2^3, \quad \{000, 011, 101, 110\}, \quad \{000, 111\}, \quad \{000\}.$$

Their inclusion lattice is



Notice that the two middle codes are incomparable:

$$111 \notin \{000, 011, 101, 110\}, \quad 110 \notin \{000, 111\}.$$

Thus the lattice has the shape of a diamond.

This example illustrates the general principle: cyclic codes correspond to the ideals of

$$\mathbb{F}_q[x]/\langle x^n - 1 \rangle,$$

and these ideals are classified by the divisors of $x^n - 1$.

Remark 9. The appearance of ideals containing $(x^n - 1)$ comes from the correspondence theorem for ideals. Indeed, cyclic codes are first identified with ideals of the quotient ring

$$\mathbb{F}_q[x]/(x^n - 1).$$

If

$$\pi : \mathbb{F}_q[x] \rightarrow \mathbb{F}_q[x]/(x^n - 1)$$

is the quotient map, then every ideal of $\mathbb{F}_q[x]/(x^n - 1)$ is of the form $J/(x^n - 1)$, where J is an ideal of $\mathbb{F}_q[x]$ satisfying

$$(x^n - 1) \subseteq J.$$

Thus the condition of containing $(x^n - 1)$ is not an additional coding-theoretic requirement; it simply records the fact that we are working modulo $x^n - 1$.

Since $\mathbb{F}_q[x]$ is a principal ideal domain, every such ideal J has the form

$$J = \langle g(x) \rangle$$

for a unique monic polynomial $g(x)$. The inclusion

$$(x^n - 1) \subseteq \langle g(x) \rangle$$

holds if and only if

$$g(x) \mid x^n - 1.$$

Hence the cyclic codes of length n over $\mathbb{F}_q$ are classified exactly by the monic divisors of $x^n - 1$.

Remark 10. Cyclic codes are ideals of the quotient ring $\mathbb{F}_q[x]/(x^n - 1)$. By the correspondence theorem, such ideals are in bijection with the ideals $J \subseteq \mathbb{F}_q[x]$ containing the kernel $(x^n - 1)$ of the quotient map

$$\mathbb{F}_q[x] \rightarrow \mathbb{F}_q[x]/(x^n - 1).$$

Because $\mathbb{F}_q[x]$ is a principal ideal domain, every such ideal has the form $(g(x))$, and the condition

$$(x^n - 1) \subseteq (g(x))$$

is equivalent to $g(x) \mid x^n - 1$. This is why cyclic codes are classified by the monic divisors of $x^n - 1$.

Remark 11 (Two complementary viewpoints on cyclic codes). There are two closely related ways to understand cyclic codes, and each serves a different purpose.

First, one may view a cyclic code $C \subseteq \mathbb{F}_q^n$ as an ideal of the quotient ring

$$\mathbb{F}_q[x]/(x^n - 1).$$

This is the more intuitive viewpoint. Indeed, under the natural $\mathbb{F}_q$ -vector space isomorphism

$$\Phi : \mathbb{F}_q^n \longrightarrow \mathbb{F}_q[x]/(x^n - 1), \quad (c_0, \dots, c_{n-1}) \longmapsto c_0 + c_1x + \dots + c_{n-1}x^{n-1},$$

cyclic shift of coordinates corresponds exactly to multiplication by x modulo $x^n - 1$. Hence the defining property of a cyclic code—namely, being an $\mathbb{F}_q$ -linear subspace closed under cyclic shift—becomes precisely the statement that $\Phi(C)$ is an ideal of $\mathbb{F}_q[x]/(x^n - 1)$.

Second, one may use the correspondence theorem for ideals. Writing

$$R = \mathbb{F}_q[x], \quad I = (x^n - 1),$$

the ideals of R/I are in one-to-one correspondence with the ideals J of R such that

$$I \subseteq J \subseteq R.$$

Thus cyclic codes correspond to ideals of $\mathbb{F}_q[x]$ containing $(x^n - 1)$. Since $\mathbb{F}_q[x]$ is a principal ideal domain, every such ideal has the form

$$J = (g(x))$$

for some polynomial $g(x)$ dividing $x^n - 1$. This is the structural viewpoint, from which the generator polynomial of a cyclic code naturally arises.

Therefore, the quotient-ring viewpoint is usually better for explaining *why* cyclic codes are related to ideals, whereas the ideal-correspondence viewpoint is better for describing *which* ideals occur and for classifying cyclic codes. In practice, the most natural presentation is to begin with the first viewpoint and then pass to the second.

A Exercises

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